SOFTWARE ENGINEERING

Lab 3 – Architecture Selection & Written Justification

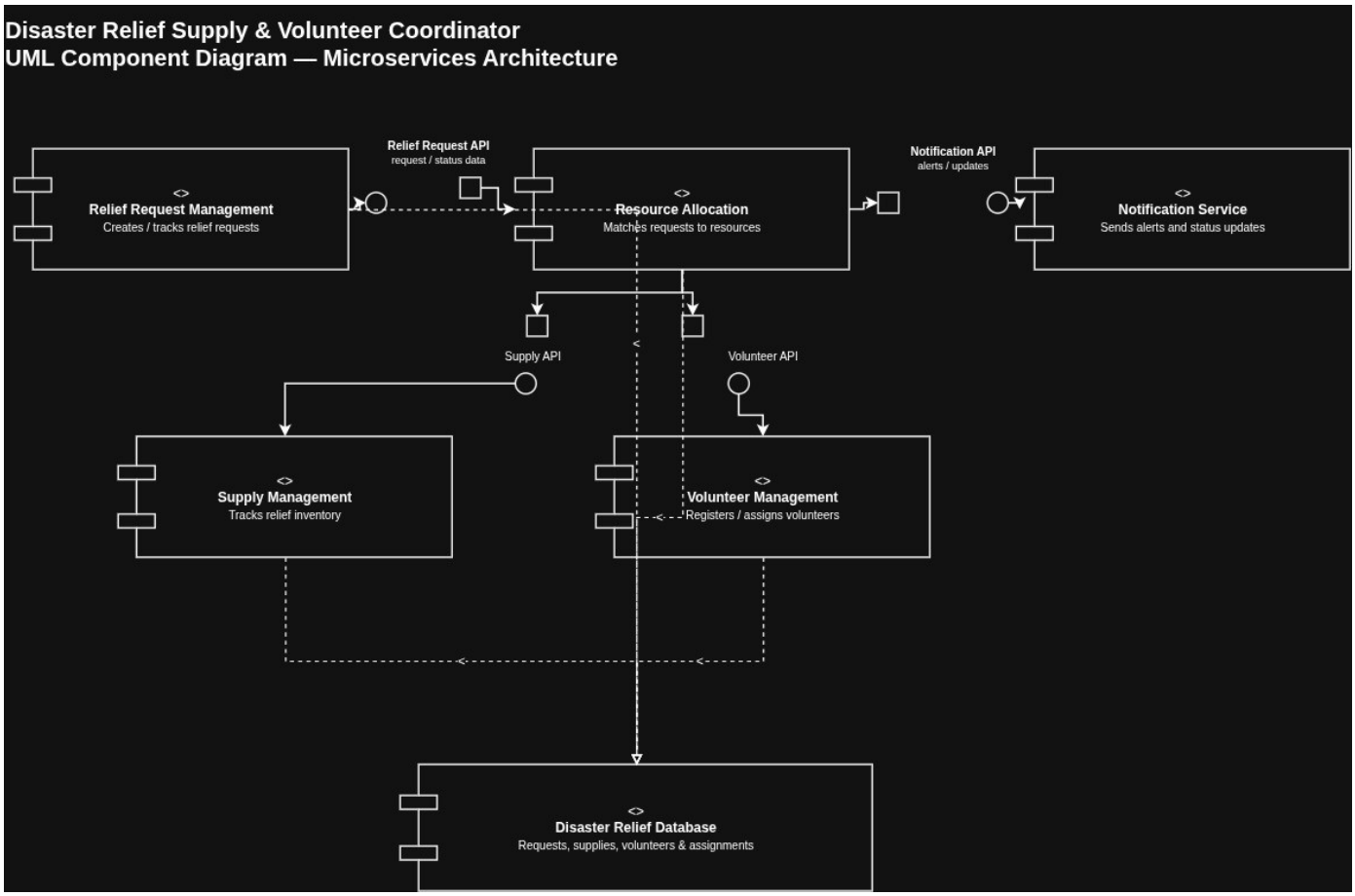
Disaster Relief Supply & Volunteer Coordinator

NAME: VASANTH KUMAR REDDY

SRN: PES2UG25CS829

SECTION:B

UML COMPONENT DIAGRAM



JUSTIFICATION

Architecture Selection

We chose Microservices Architecture for the Disaster Relief Supply & Volunteer Coordinator System.

Architectural Choice

The system is divided into independently deployable services for relief requests, supply management, volunteer management, resource allocation and notifications, with a shared data component represented by the Disaster Relief Database.

Reason 1 – Independent Scaling

Disaster situations can create uneven workloads. Relief-request processing may receive a sudden surge of requests while other services receive fewer requests. Microservices allows heavily used services to be scaled independently instead of scaling the entire application.

Reason 2 – Fault Isolation

Disaster-relief coordination requires resilience. If a supporting service such as notifications becomes unavailable, core services such as supply and volunteer management can continue operating. Separating services therefore limits the impact of individual failures.

Security Advantage

Service boundaries allow authentication and authorization to be applied to individual APIs. Access to sensitive relief-request, volunteer and allocation information can therefore be restricted to the services and users that require it.

Performance Benefit

Independent scaling allows high-demand services, such as relief-request and supply-management services, to receive additional resources without unnecessarily scaling the rest of the system. This can improve responsiveness during demand spikes.

Design Note

The supplied Lab 3 document specifies the architectural styles, UML notation and deliverable format, but its detailed example requirements are for a Coffee Kiosk. The components above are therefore a scenario-based design for the assigned Disaster Relief scenario.